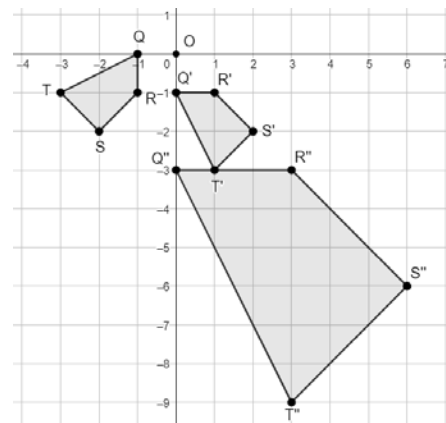


A sequence of transformations of $QRST$ is shown on the coordinate grid. $QRST$ is rotated then dilated, where $Q'R'S'T'$ is the rotation and $Q''R''S''T''$ is the dilation of $Q'R'S'T'$.



Complete the descriptions:

- $QRST$ was _____ to create $Q'R'S'T'$.
- $Q'R'S'T'$ was dilated using a scale factor of _____ centered at _____ to create $Q''R''S''T''$.
- Complete the table:

Transformation	Algebraic Description
Rotation	$(x,y) \rightarrow$
Dilation	$(x,y) \rightarrow$

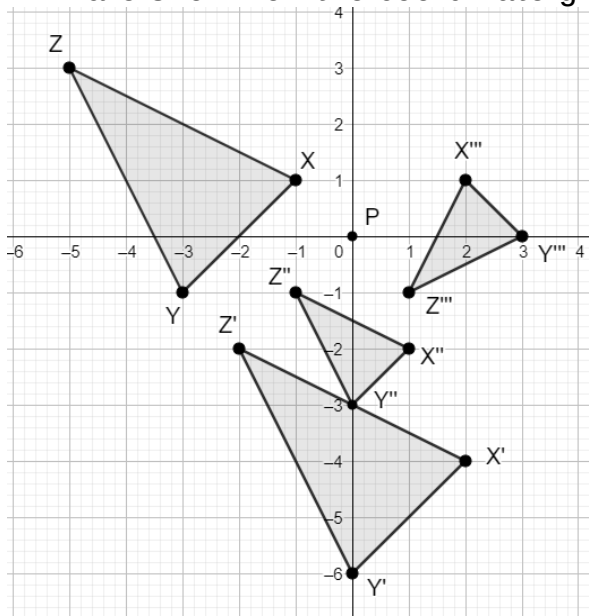
- Determine the coordinates if $QRST$ is first dilated then rotated:

Preimage $QRST$		Dilation $Q'R'S'T'$		Rotation $Q''R''S''T''$	
Q	$(-1,0)$	Q'		Q''	
R	$(-1,-1)$	R'		R''	
S	$(-2,-2)$	S'		S''	
T	$(-3,-1)$	T'		T''	

- Based on the results above, the sequence of transformations commutative.

o is
o is not

$XYZ, X'Y'Z', X''Y''Z'',$ and $X'''Y'''Z'''$ are shown on the coordinate grid.



6. Determine what transformations occurred and in what order:
- XYZ is first

☐ translated
☐ reflected
☐ rotated
☐ dilated

 to $X'Y'Z'$, then

☐ translated
☐ reflected
☐ rotated
☐ dilated

 to $X''Y''Z''$,
- and finally

☐ translated
☐ reflected
☐ rotated
☐ dilated

 to $X'''Y'''Z'''$.
7. Describe the transformation for $XYZ \rightarrow X'Y'Z'$.
8. Describe the transformation for $X'Y'Z' \rightarrow X''Y''Z''$.
9. Describe the transformation for $X''Y''Z'' \rightarrow X'''Y'''Z'''$.

10. Write the algebraic description for the sequence of transformations from 7-9:

Transformation	Algebraic Description
	$(x,y) \rightarrow$
	$(x,y) \rightarrow$
	$(x,y) \rightarrow$